

Project Development Phase

Data Preprocessing

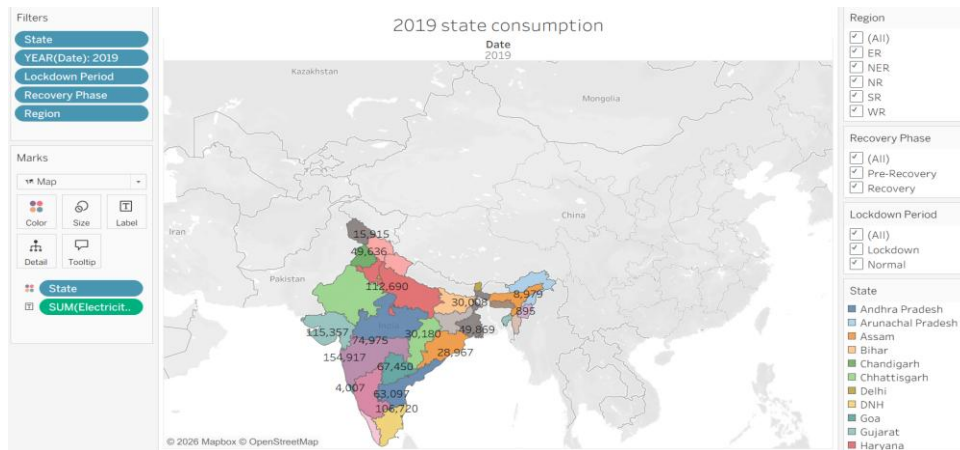
The electricity consumption dataset was preprocessed in Tableau Desktop to ensure data quality, consistency, and accuracy before creating the visualizations. The following preprocessing steps were performed:

1. Imported the electricity consumption dataset (CSV file) into Tableau Desktop.
 2. Verified the dataset for completeness and correctness.
 3. Checked for missing values and handled them appropriately.
 4. Removed duplicate records wherever applicable.
 5. Renamed column headers for better readability and consistency.
 6. Verified and corrected data types for all fields, including **Date, State, Region, Quarter, and Electricity Consumption**.
 7. Converted the **Date** field into the appropriate date format for time-based analysis.
 8. Created calculated fields required for analysis and visualization.
 9. Sorted and organized the data to improve clarity in visualizations.
 10. Applied appropriate formatting to labels, titles, legends, tooltips, and color schemes.
 11. Applied filters such as **Year, State, Region, Quarter, and Recovery Phase** to support data analysis and visualization.
 12. Validated the processed dataset to ensure accurate and reliable analysis before creating the visualizations.
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Business Questions with Visualizations

Business Question 1

How was electricity consumption distributed across different states in 2019?



Visualization Used: Filled Map (2019)

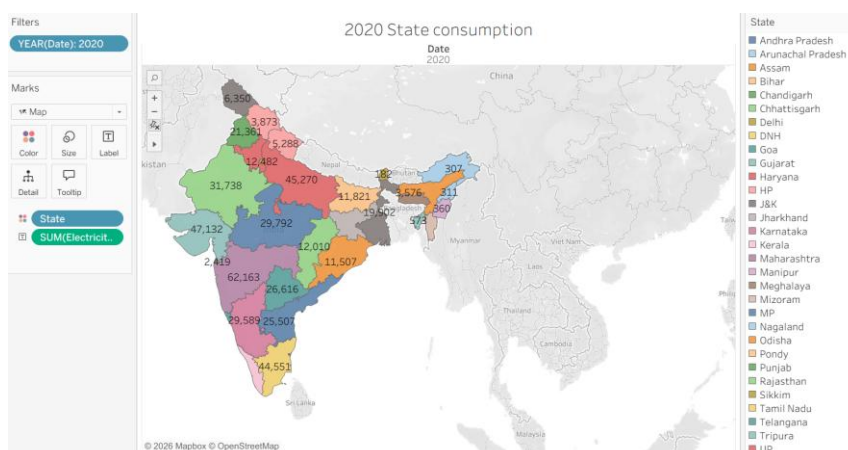
Insight:

The filled map displays state-wise electricity consumption for 2019, highlighting variations in electricity usage across different states. And observed that Maharastra state has a highest Electricity Consumption in 2019

Business Question 2

How was electricity consumption distributed across different states in 2020?

Visualization Used: Filled Map (2020)



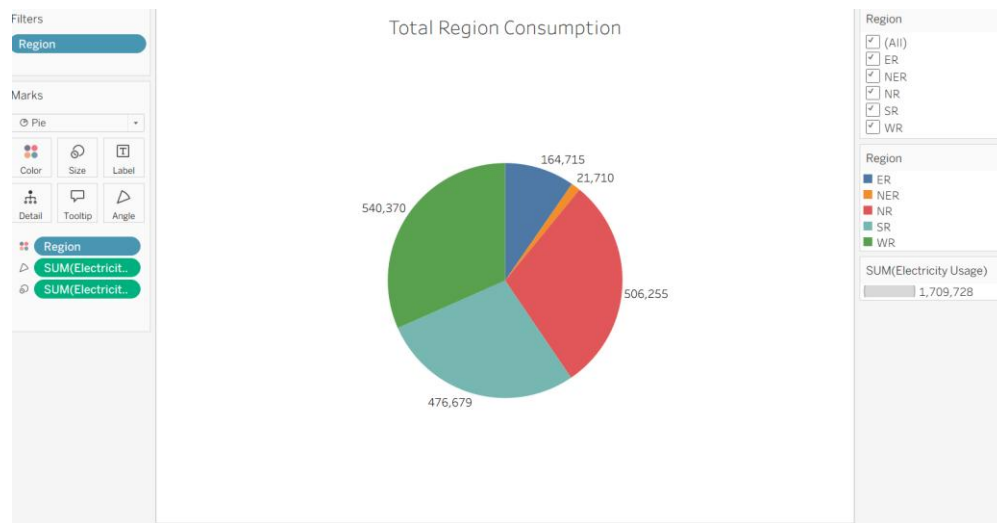
Insight:

The visualization illustrates state-wise electricity consumption for 2020 and enables comparison with the previous year's consumption pattern. Maharastra has a highest Electricity usage in 2020

Business Question 3

Which region contributed the highest electricity consumption?

Visualization Used: Pie Chart



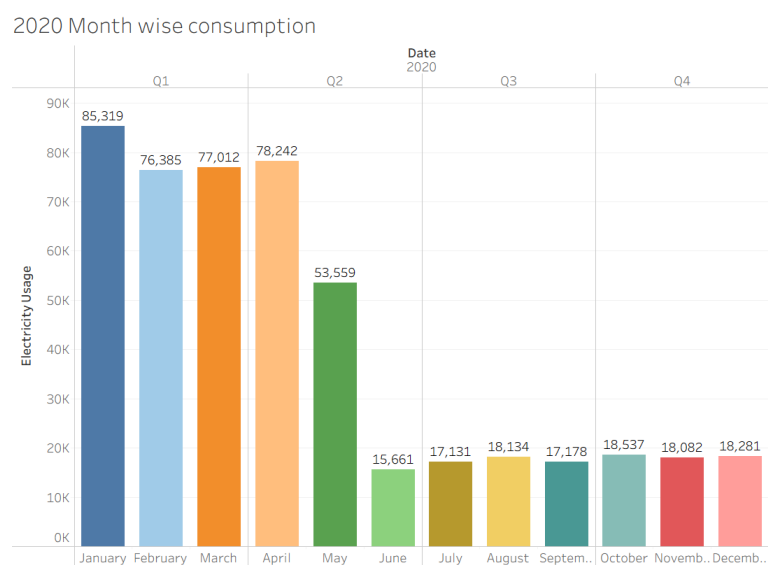
Insight:

The pie chart compares electricity consumption across different regions and shows each region's contribution to the total electricity consumption. North region has highest Electricity Usage

Business Question 4

How did electricity consumption vary across different quarters?

Visualization Used: Bar Chart



Insight:

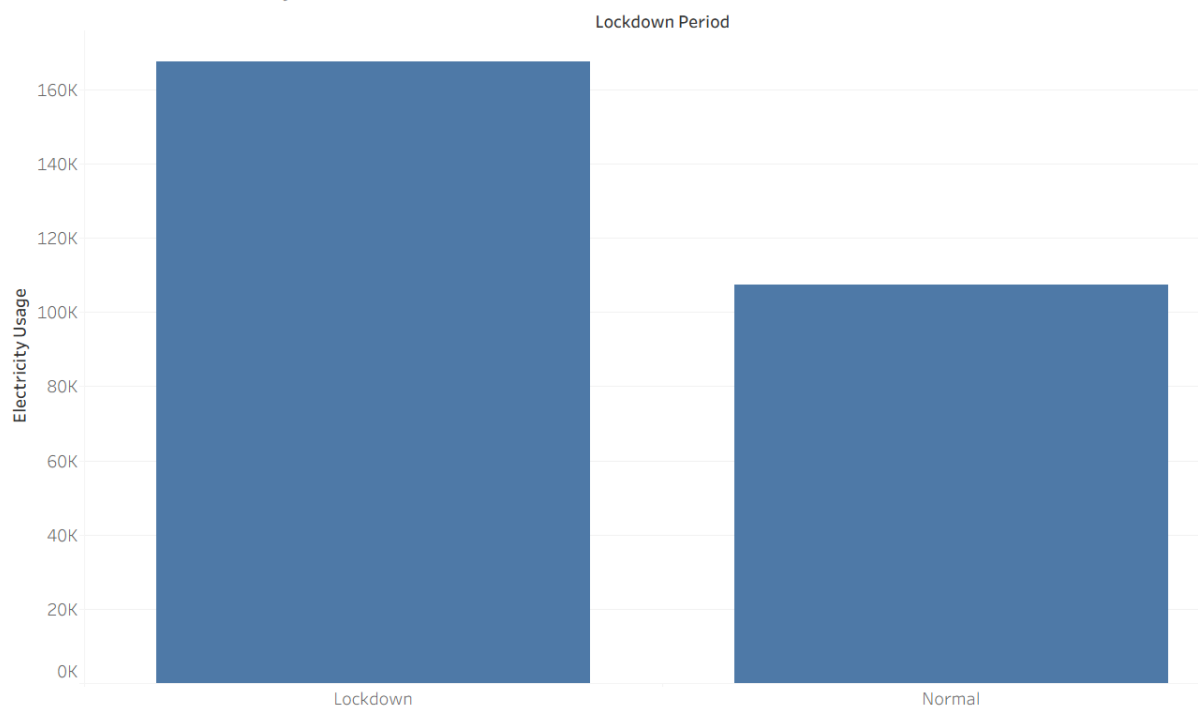
The bar chart compares electricity consumption across different quarters and helps identify seasonal consumption patterns. January and February has highest Electricity Consumption in 2020

Business Question 5

How did electricity consumption change during the COVID-19 lockdown and recovery phases?

Visualization Used: Bar Chart

Lockdown vs Recovery

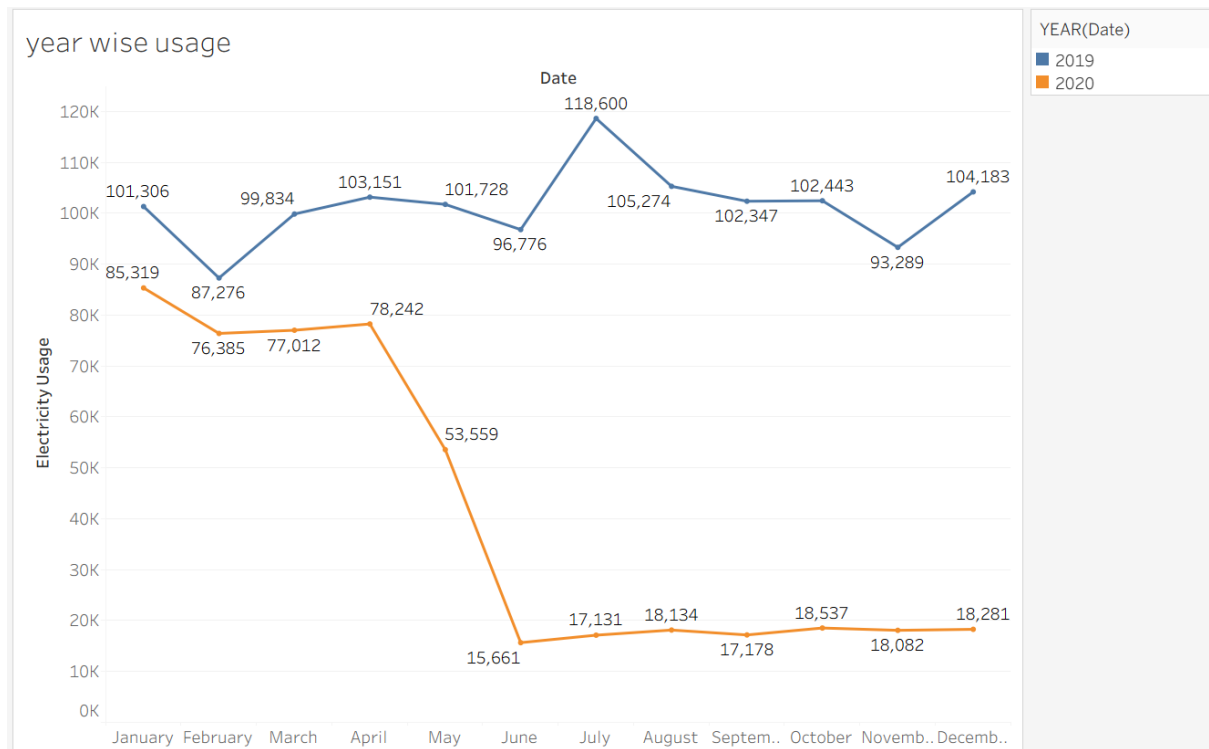
**Insight:**

The visualization compares electricity consumption during the lockdown and recovery periods, highlighting changes in electricity demand. Electricity Consumption is more Highest in Lockdown than Normal

Business Question 6

How has electricity consumption changed over the years?

Visualization Used: Line Chart



Insight:

The line chart illustrates yearly electricity consumption trends and helps identify changes in electricity demand over time.